Assignment 7

Due: 3/25

Note: Show all your work.

Problem 1 (20 points). Consider the following transactional database.

TID	Items
100	2, 4, 5, 6
200	1, 4, 5, 7
300	2, 4, 5
400	1, 2, 4, 5, 6, 7
500	1, 2, 6

(1) Mine all frequent itemsets using the Apriori algorithm that we discussed in the class. Show all candidate itemsets and frequent itemsets. You should follow the step by step process that we discussed in the class (i.e., C1 → L1 → C2 → L2 → …). You don't need to show the pruning steps. Minimum support = 30% (or 2 or more transactions). To save your time, L1 is given below:

L1:

Itemset	1	2	4	5	6	7
Count	3	4	4	4	3	2

(2) Sort all frequent 4-itemsets by their item number. Then, select the first frequent 4-itemset from the sorted list of frequent 4-itemsets and mine all strong rules from this itemset that have the format {W, X} => {Y, Z}, where W, X, Y, and Z are individual items. Assume that minimum confidence = 80%.

Problem 2 (10 points). Consider the following contingency table.

	C (buys coffee = Yes)	$\overline{C}$ (buys coffee = No)
T (buys tea = Yes)	264	75
$\overline{T}$ (buys tea = No)	48	132

Compute the *lift*, *all-confidence*, *cosine*, *Kulczynski* and *imbalance ratio* measure, and determine whether buying coffee and buying tea are positively correlated, negatively correlated, or not correlated.

Problem 3 (20 points). This problem is about mining frequent itemsets and strong rules from a grocery data using R. Use *groceries3000.csv* data for this problem. It is a slightly modified version of the *groceries.csv* data that is included in R.

(1). Mine all frequent itemsets with the minimum support = 4% and include the screenshot of the summary in your submission.
(2). What is the total number of frequent itemsets? List the top five frequent itemsets and their supports in descending order of support.

For the following 4 questions, if any of *Li* is empty, then answer "zero."

(3). What is the total number of frequent 1-itemsets (L1)? List the top five frequent 1-itemsets and their supports in descending order of support.
(4). What is the total number of frequent 2-itemsets (L2)? List the top five frequent 2-itemsets and their supports in descending order of support.
(5). What is the total number of frequent 3-itemsets (L3)? Show the frequent 3-itemset and its support that has the highest support.
(6). What is the total number of frequent 4-itemsets (L4)? Show the frequent 4-itemset and its support that has the highest support.

(7). Mine all rules with minimum support = 1% and minimum confidence = 25% and include the screenshot of the summary in your submission.
(8). List the top five rules and their confidences in descending order of confidence.
(9). List the top five rules and their supports in descending order of support.
(10). Show all rules that include *yogurt* (in either side of a rule) along with their supports and confidences.

Submission:

Submit all solutions in a single Word or PDF document and upload it to Blackboard. Use *LastName_FirstName_hw7*.docx or *LastName_FirstName_hw7*.pdf as the file name. You must also submit your R code file. Name it *hw7.R*.